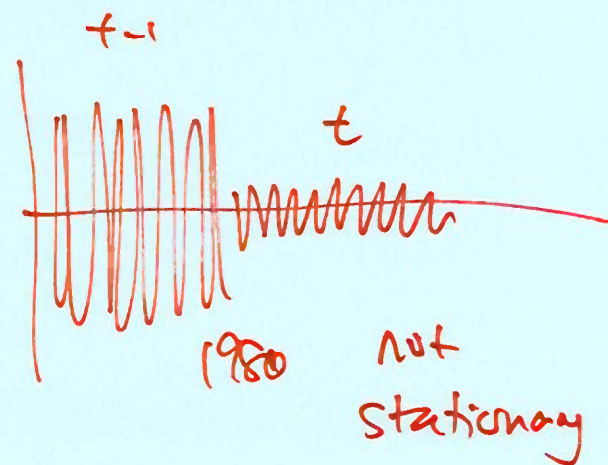


$$y_t = \phi_0 + \textcircled{u_t} + \theta_1 u_{t-1}$$

whaaaaat.

$$u_t \sim \text{iid} (0, \sigma_u^2)$$

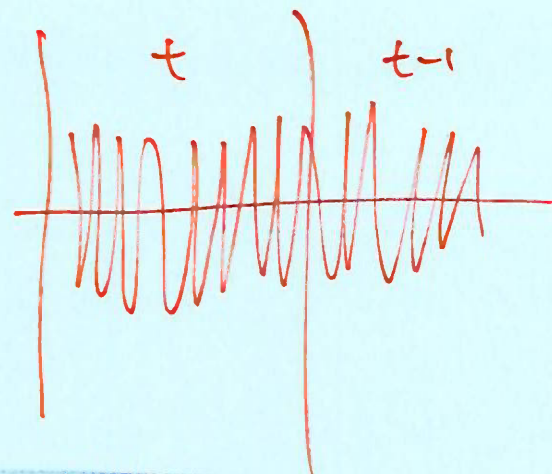


$$E(u_t u_t) = E(u_t^2) = \sigma_u^2 \quad \text{var.}$$

$$E(u_{t-1} u_{t-1}) = E(u_{t-1}^2) = \sigma_u^2$$

$$E(u_{t-2} u_{t-2}) = E(u_{t-2}^2) = \sigma_u^2$$

Stationary.



$$E(u_t u_{t-1}) = 0$$

$\Rightarrow$ "i" means not linearly predictable from past info.

$$E(u_t u_{t-2}) = 0$$

$$E(u_{t-1} u_{t-2}) = 0$$

do not

$$E(u_t u_{t-3}) = 0 \quad \text{not predictable.}$$

slide 39

$$\theta_1 = 0.5$$

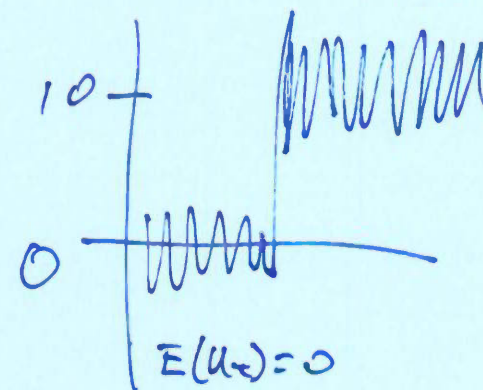
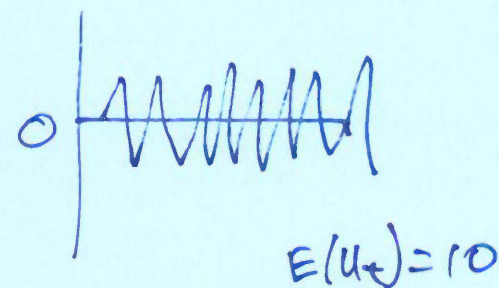
$$\begin{aligned} E(y_t) &= E(\phi_0 + u_t + \theta_1 u_{t-1}) \\ &= E(\phi_0) + E(u_t) + \overbrace{E(\theta_1 u_{t-1})} \\ &= \phi_0 + 0 + \theta_1 E(u_{t-1}) \\ &= \phi_0 + \theta_1 E(u_t) \\ &= \phi_0 + 0 \text{ done.} \\ &= \mu \uparrow \end{aligned}$$

$$\begin{aligned} \mu &= \phi_0 \\ \downarrow \\ \mu & \end{aligned}$$

$$\phi_0 = \text{constant}$$

$$u_t \sim \text{iid}(0, \sigma_u^2)$$

$\downarrow$
 $E(u_t)$



Slide 41

$$Y_t - \mu = u_t + \theta_1 u_{t-1} \Rightarrow Y_{t-1} - \mu = u_{t-1} + \theta_1 u_{t-2}$$

$$\downarrow$$

$$\phi_0 \Rightarrow Y_{t-2} - \mu = u_{t-2} + \theta_1 u_{t-3}$$

$$\sigma_0 = E(Y_t - \mu)^2 \Rightarrow \text{variance}.$$

$$= E(u_t + \theta_1 u_{t-1})^2$$

$$= E(u_t^2 + 2u_t \theta_1 u_{t-1} + \theta_1^2 u_{t-1}^2)$$

$$= E(u_t^2) + 2E(u_t \theta_1 u_{t-1}) + E(\theta_1^2 u_{t-1}^2)$$

$$= \sigma_u^2 + 2\theta_1 E(u_t u_{t-1}) + \theta_1^2 E(u_{t-1}^2)$$

$$= \sigma_u^2 + \theta_1^2 E(u_t^2)$$

" 0

stationary

$$= (1 + \theta_1^2) \sigma_u^2 \rightarrow \text{slide 42}$$

Slide 43

(First-order autocovariance)

$\sigma_0 = \text{variance}$
↑

$$\gamma_1 = E(Y_t - \mu)(Y_{t-1} - \mu)$$

love = 0

$$= E(u_t + \theta_1 u_{t-1})(u_{t-1} + \theta_1 u_{t-2})$$

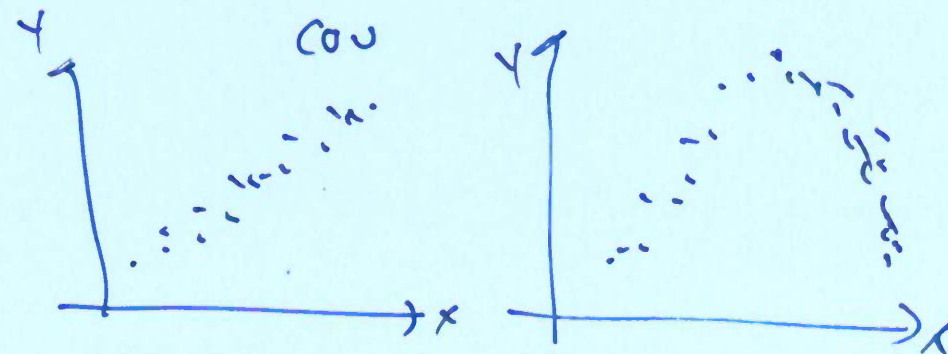
$$= E(u_t u_{t-1} + \theta_1 u_t u_{t-2} + \theta_1 u_{t-1}^2 + \theta_1^2 u_{t-1} u_{t-2})$$

$$= E(u_t u_{t-1}) + \theta_1 E(u_t u_{t-2}) + \theta_1 E(u_{t-1}^2) + \theta_1^2 E(u_{t-1} u_{t-2})$$

$$= 0 + 0 + \theta_1 E(u_t^2) + 0$$

↓
stationary

$$= \theta_1 \sigma_u^2$$



Slide 46

First-order autocorrelation $\rho_1 = \frac{\sigma_1}{\sigma_0}$ unit free

$$-1 \leq \rho_1 \leq 1$$

hope. ρ_1 closer to 1 $\Rightarrow$ forecast with better accuracy.

$$\rho_1 = \frac{\sigma_1}{\sigma_0} = \frac{\theta_1 \sigma_u^2}{(1 + \theta_1^2) \sigma_u^2} = \frac{\theta_1}{1 + \theta_1^2} \quad \text{Slide 46}$$

$\neq 0$

$$\begin{array}{ccc} \rho_2 = \frac{\sigma_2}{\sigma_0} = 0 & \rho_3 = 0 & \rho_4 = 0 \dots \dots \end{array}$$

$\downarrow \qquad \qquad \uparrow \qquad \qquad \uparrow$

Second-order autocorrelation

Slide 45

$$\gamma_2 = E(Y_t - \mu)(Y_{t-2} - \mu)$$

$$= E(u_t + \theta_1 u_{t-1})(u_{t-2} + \theta_1 u_{t-3})$$

$$\Rightarrow = E(u_t u_{t-2} + \theta_1 u_t u_{t-3} + \theta_1 u_{t-1} u_{t-2} + \theta_1^2 u_{t-1} u_{t-3})$$

$$= E(u_t u_{t-2}) + \theta_1 E(u_t u_{t-3}) + \theta_1 E(u_{t-1} u_{t-2}) + \theta_1^2 E(u_{t-1} u_{t-3})$$

0 + 0 + 0 + 0

$$= 0$$

|:0|

|:06 back.

$$\rho_2 = \frac{\gamma_2}{\gamma_0} = \frac{0}{\gamma_0} = 0$$

we slide 42

$$E(u_t | F_{t-1}) = 0 \quad \text{MDS}$$


gap = 1

$$E(u_t^2 | F_{t-1}) = 1 \quad \text{unit variance.}$$


gap = 1

$$E(u_{t+2} | F_t) \quad \text{not MDS}$$


gap is not one.

$$E(u_t | F_{t-10000}) \quad \text{not MDS}$$

$$E(u_{t+10000}^2 | F_{t-1}) = \text{don't know}$$


gap is not one.

$$F = I$$

To show:

slide 4

$$E(\varepsilon_{t+1}^2 | F_t) = E(\sigma_{t+1}^2 | F_t)$$

replace with

slide 4 $\Rightarrow h=2$

$$E(\varepsilon_{t+1}^2 | F_t) = E\left(E(\varepsilon_{t+1}^2 | F_t) \mid F_t\right)$$

gap 2

$$E(\varepsilon_{t+2}^2 | F_t) = E(\sigma_{t+2}^2 | F_t)$$

$$E(\varepsilon_{t+2}^2 | F_t) = E\left(E(\varepsilon_{t+2}^2 | F_{t+1}) \mid F_t\right)$$

replace with

gap = 1

slide 4

$$E(\varepsilon_{t+3}^2 | F_t) = E(\sigma_{t+3}^2 | F_t)$$

$$E(\varepsilon_{t+3}^2 | F_t) = E\left(E(\varepsilon_{t+3}^2 | F_{t+2}) | F_t \right)$$

gap = 1
not gap = -1

$$E(\varepsilon_{t+1}^2 | F_t) = E(E(\varepsilon_{t+1}^2 | F_t) | F_t)$$

$$\varepsilon_{t+1} = \sigma_{t+1} u_{t+1} \Leftrightarrow \varepsilon_{t+1}^2 = \sigma_{t+1}^2 u_{t+1}^2$$

$$= E(E(\cancel{\varepsilon_{t+1}^2} \sigma_{t+1}^2 u_{t+1}^2 | F_t) | F_t)$$

$$\sigma_{t+1}^2 = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2 \Rightarrow \text{GARCH}(1,1)$$

$\checkmark$ ^a number = is known given F_t why $\varepsilon_t^2, \sigma_t^2$ is known today

$$= E(\sigma_{t+1}^2 E(u_{t+1}^2 | F_t) | F_t)$$

$$= E(\sigma_{t+1}^2 | F_t)$$

replace with

Slide 4 to 6

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \quad \# \text{ GARCH (1,1)}$$

One-Step-ahead forecast $\Rightarrow$ slide 7

$$E(\sigma_{t+1}^2 | F_t) = E(\alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2 | F_t)$$

best guess of σ_{t+1}^2 having obs info up until F_t

$$= E(\alpha_0 | F_t) + E(\alpha_1 \varepsilon_t^2 | F_t) + E(\beta_1 \sigma_t^2 | F_t)$$

$$= \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2$$

= slide 7.

two-step-ahead forecast:

$$\begin{aligned} E(\sigma_{t+2}^2 | F_t) &= E(\alpha_0 + \alpha_1 \varepsilon_{t+1}^2 + \beta_1 \sigma_{t+1}^2 | F_t) \\ &= E(\alpha_0 | F_t) + E(\alpha_1 \varepsilon_{t+1}^2 | F_t) + \beta_1 E(\sigma_{t+1}^2 | F_t) \\ &= \alpha_0 + \alpha_1 E(\varepsilon_{t+1}^2 | F_t) + \beta_1 E(\sigma_{t+1}^2 | F_t) \\ &= \alpha_0 + \alpha_1 E(\sigma_{t+1}^2 | F_t) + \beta_1 E(\sigma_{t+1}^2 | F_t) \\ &= \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+1}^2 | F_t). \end{aligned}$$

1-step ahead forecast.

Slide 8